



# Mitoprep Answer Key 6

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**Q1) Explanation:**

**Hint:** loss in P.E. is equal to the sum of elastic potential energy stored and heat energy.

**Step:** Analyse each statement one by one.

A body of mass  $M$  is attached to the lower end of the metal wire and after the elongation of the wire is shown in the figure below;



The loss in gravitational potential energy of the system is  $Mgl$ .

The electric potential energy stored in the wire is given by;

$$P.E = \frac{1}{2} \times \text{Stress} \times \text{Strain} \times V$$

$$\Rightarrow P.E = \frac{1}{2} \times \frac{Mg}{A} \times \frac{\Delta L}{L} \times AL$$

$$\Rightarrow P.E = \frac{1}{2} Mg \Delta L = \frac{1}{2} Mgl \quad [\Delta L = l]$$

Half of the potential energy is stored as elastic potential energy and the remaining half is dissipated as heat.

So, the heat produced is  $\frac{1}{2} Mgl$

Therefore, the correct statement is (A), (C), and (D) only.

Hence, option (4) is the correct answer.

**Q2) Explanation:**

**Hint:** Find pressure difference.

**Step 1:** Find the pressure difference.

$$P_0 = 760 \text{ mm Hg}$$

$$P = 750 \text{ mm Hg}$$

$$\Delta P = 10 \text{ mm Hg}$$

**Step 2:** Find the maximum depth.

$$10 \text{ mm Hg} = h \text{ cm of water}$$

$$\rho_{Hg}(l)g = \rho_w(h)g$$

$$h = 13.6 \text{ cm}$$

Hence, option (3) is the correct answer.

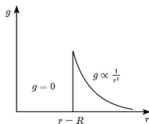
**Q3) Explanation:**

**Hint:** For the spherical shell, the gravitational field at the centre is zero.

**Explanation:** In the case of the spherical shell, the gravitational field inside the shell is zero, and outside the shell, the gravitational field decreases as a rectangular hyperbola as shown in the figure below.

$$g_{in} = 0 \text{ for } r \leq R$$

$$g_{out} = \frac{GM}{r^2} \text{ for } r > R$$



Hence, option (1) is the correct answer.

#### Q4) Explanation:

**Hint:** The acceleration due to gravity at the centre of the earth is zero.

**Step:** Find the variation of acceleration due to gravity as a function of  $r$ .

The variation of acceleration due to gravity as a function of  $r$  are given as below;

$$\Rightarrow g \propto r \quad \text{for } r < R$$

$$\Rightarrow g \propto \frac{1}{r^2} \quad \text{for } r > R$$

Therefore, the variation of acceleration due to gravity as a function of  $r$  is shown in the figure below;



Hence, option (1) is the correct answer.

#### Q5) Explanation:

**Hint:**  $E = T \times A$

**Step:** Analyse each statement one by one.

In merging the small droplets to form a single drop, some energy is released in the process.

Therefore, statement (A) is correct.

The initial surface energy of the  $n$  small droplets is  $nE$ .

As the volume of the system remains constant i.e.,  $n \times \frac{4}{3}\pi r^3 = \frac{4}{3}\pi R^3$

$$\Rightarrow R = n^{\frac{1}{3}}r$$

The final surface energy of the drop is given by;

$$E' = (4\pi R^2)T = (4\pi(n^{\frac{1}{3}}r)^2)T$$

$$\Rightarrow E' = n^{\frac{2}{3}} \times (4\pi r^2)T = n^{\frac{2}{3}}E$$

The energy released or absorbed will be given by;

$$\Delta E = nE - E' = nE - n^{\frac{2}{3}}E$$

$$\Rightarrow \Delta E = E \left( n - n^{\frac{2}{3}} \right)$$

Therefore, the correct statements are (A) and (C) only.

Hence, option (1) is the correct answer.

#### Q6) Explanation:

**Hint:**  $\theta = \omega_0 t + \frac{1}{2}\alpha t^2$ .

**Step 1:** Find the angular acceleration of the wheel.

In the first second ( $t = 1$ ), the angle rotated is given as:

$$\theta_1 = 2 \text{ radians}$$

The angular acceleration of the block is given by;

$$\theta = \omega_0 t + \frac{1}{2}\alpha t^2$$

$$\Rightarrow 2 = \frac{1}{2}\alpha t^2 \quad [\omega_0 = 0]$$

$$\Rightarrow 2 = \frac{\alpha}{2}$$

$$\Rightarrow \alpha = 4 \text{ rad/s}^2$$

**Step 2:** Find the angle rotated in the next second.

$$\theta = \omega_0 t + \frac{1}{2}\alpha t^2$$

$$\Rightarrow \theta_2 = \frac{1}{2} \times 4 \times 2^2 = 8 \text{ radian}$$

The angle turned during the  $2^{\text{nd}}$  second is  $(8 - 2)$  radian = 6 radians

Therefore, the angle through which the wheel rotated in the next second is 6 radians.

Hence, option (3) is the correct answer.

.Q7) As we know Young's modulus is  $Y = \frac{F l}{A \Delta l}$

$$\text{Since, } V = A l \Rightarrow l = \frac{V}{A}$$

So,

$$F = \frac{Y A \Delta l}{l} = \frac{Y A^2 \Delta l}{V}$$

For the same volume and the same amount of extension :  $F \propto A^2$

$$\frac{F_1}{F_2} = \left( \frac{A_1}{A_2} \right)^2$$

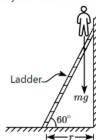
$$\frac{F_1}{F_2} = \left( \frac{A}{3A} \right)^2 = \frac{1}{9}$$

$$F_2 = 9F$$

Q8) Explanation:

**Hint:** The net torque increases as a person climbs on the ladder.

**Explanation:** When a person climbs a ladder, he exerts a force on the ladder that is directed upwards. This force tends to tip the ladder over, causing it to slip. The closer the person is to the top of the ladder, the greater the torque (twisting force) they exert on the ladder, and the more likely the ladder is to slip.



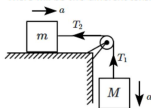
Hence, option(4) is the correct answer.

Q9) Explanation:

**Hint:**  $a = r\alpha$

**Step 1:** Identify the tension in the system.

There will be two different tensions in both strings named  $T_1$  and  $T_2$



**Step 2:** Write the force equation for the blocks.

$$Mg - T_1 = Ma \quad (1)$$

$$T_2 = ma \quad (2)$$

**Step 3:** Write torque equation for Pulley.

$$(T_1 - T_2)r = I\alpha \quad (3)$$

No slipping condition:  $a = r\alpha \quad (4)$

**Step 4:** Find acceleration.

using equation (1), (2) and (4) in equation (3) we get

$$Mg - Ma - ma = I \frac{a}{r^2}$$

$$(M + m + \frac{I}{r^2})a = Mg$$

$$a = \frac{Mg}{M + m + \frac{I}{r^2}}$$



**Q10) Explanation:**

**Hint:**  $Q = Av$

**Step:** Find the speed of the blood flow in the combined vein.

The volume flow rate through the vein is given by;  $Q = Av$

$$\Rightarrow \pi r^2 v + \pi r^2 v = \pi (2r)^2 v'$$

$$\Rightarrow 2\pi r^2 v = \pi 4r^2 v'$$

$$\Rightarrow v' = \frac{v}{2}$$

Therefore, the speed of the blood flow in the combined vein is  $\frac{v}{2}$ .

Hence, option (2) is the correct answer.

**Q11) Explanation:**

**Hint:** Apply Bernoulli's theorem.

**Step:** Find the pressure difference across the wings.

$$P_1 + \frac{1}{2} \rho v_1^2 = P_2 + \frac{1}{2} \rho v_2^2$$

$$P_1 - P_2 = \frac{1}{2} \rho (v_2^2 - v_1^2) = \frac{1}{2} \rho (v_2 + v_1) (v_2 - v_1)$$

$$= \frac{1}{2} \times 1.3(120 + 90)(120 - 90)$$

$$= \frac{1}{2} \times 1.3 \times 210 \times 30 = 4095 \text{ N/m}^2$$

Hence, option (4) is the correct answer.

**Q12) Hint:**  $\Delta E = U_2 - U_1$

**Step 1:** Find the total energy of orbiting satellite.

$$U = -\frac{GM_s m}{2r}$$

**Step 2:** Find the energy of the satellite in orbital radius  $r$  and  $2r$ .

$$U_i = -\frac{GM_s m}{2r}$$

$$U_f = -\frac{GM_s m}{4r}$$

$$E = U_f - U_i$$

$$\Rightarrow E = \frac{GM_s m}{4r}$$

**Step 3:** Find the energy required to transfer the satellite to radius  $3r$ .

$$U_{3r} = -\frac{GM_s m}{6r}$$

$$E' = U_{3r} - U_{2r} = \frac{GM_s m}{12r} = \frac{E}{3}$$

**Q13) Explanation:**

**Hint:** Capillary rise,  $h = \frac{2T \cos \theta}{\rho g r}$

**Step 1:** Find the relation between the angles.

The height of the capillary rise is given by;

$$h = \frac{2T \cos \theta}{\rho g r}$$

Since the liquids rise to the same height  $h$  in identical capillaries, the equation becomes:

$$h = \frac{2T \cos \theta_1}{\rho_1 g r} = \frac{2T \cos \theta_2}{\rho_2 g r} = \frac{2T \cos \theta_3}{\rho_3 g r}$$

For each liquid, we can solve for the contact angle as:

$$\cos \theta_1 \propto \rho_1, \quad \cos \theta_2 \propto \rho_2, \quad \cos \theta_3 \propto \rho_3$$

As given in the question:  $\rho_1 > \rho_2 > \rho_3$  then

$$\cos \theta_1 > \cos \theta_2 > \cos \theta_3$$

As the cosine function decreases as the angle increases then,

$$\theta_1 < \theta_2 < \theta_3$$

**Step 2:** Relate the angles of contact.

As we have the relation between the cosine angles as  $\cos \theta_1 > \cos \theta_2 > \cos \theta_3$  and the angles are all less than or equal to  $\frac{\pi}{2}$  (since we are dealing with capillary rise, where angles are typically acute).

Therefore, the correct angle of contact will follow the relation;

$$0 \leq \theta_1 < \theta_2 < \theta_3 \leq \frac{\pi}{2}.$$

Hence, option (2) is the correct answer.

**Q14) Explanation:**

**Hint:**  $a = r\alpha$

**Step:** Find the acceleration of the rope.

Given:  $F = 40 \text{ N}$ ,  $r = 0.5 \text{ m}$ ,  $\tau_{\text{friction}} = 10 \text{ N-m}$ ,  $I = 0.5 \text{ kg-m}^2$

The net torque acting on the pulley is given by;

$$\tau_{\text{net}} = F \times r - \tau_{\text{friction}}$$

$$\Rightarrow \tau_{\text{net}} = 40 \times 0.5 - 10 = 10 \text{ N-m}$$

The net torque acting on the pulley is given by;

$$\tau_{\text{net}} = I\alpha$$

$$\Rightarrow \alpha = \frac{\tau_{\text{net}}}{I} = \frac{10}{0.5} = 20 \text{ rad/s}^2$$

As the rope does not slip on the pulley then,  $a = \alpha r$

$$\Rightarrow a = 20 \times 0.5 = 10 \text{ m/s}^2$$

Hence, option (1) is the correct answer.

**Q15) Explanation:**

**Hint:**  $E \propto A$

**Step 1:** Find the radius of the smaller droplets.

Let the radius of the small droplet is  $r'$  m

A liquid droplet of radius  $r$  breaks up into  $N$  identical droplets then

$$N \times \frac{4}{3} \pi r'^3 = \text{constant} \Rightarrow r' \propto \frac{1}{N^{\frac{1}{3}}} \dots (1)$$

**Step 2:** Find the total surface energy of the smaller droplets.

The total surface energy of the smaller droplets is given by;  $E \propto N \times r'^2 \dots (2)$

$$\text{From equations (1) and (2) we get; } E \propto N \times \frac{1}{N^{\frac{2}{3}}} \Rightarrow E \propto N^{\frac{1}{3}}$$

Therefore, the total surface energy of the smaller droplet is proportional to  $N^{\frac{1}{3}}$

Hence, option (3) is the correct answer.

**Q16) Explanation:**

**Hint:**  $P + \frac{1}{2}\rho v^2 + \rho gh = \text{constant}$

**Step:** Find the force exerted by the wind on the roof.

From Bernoulli's theorem,

$$P_1 + \frac{1}{2}\rho v_1^2 = P_2 + \frac{1}{2}\rho v_2^2$$

where,  $P_1, P_2$  are the pressures inside and outside the roof, and  $v_1, v_2$  are the velocities of wind inside and outside the roof. Neglect the width of the roof.

The pressure difference is given by;

$$P_1 - P_2 = \frac{1}{2}\rho v_2^2 - \frac{1}{2}\rho v_1^2$$

$$\Delta P = 960 \text{ N/m}^2$$

The force acting on the roof is given by;

$$F = (P_1 - P_2) A = 960 \times 250$$

$$F = 24 \times 10^4 \text{ N} = 2.4 \times 10^5 \text{ N}$$

As the pressure inside the roof is more than outside to it. So the force will act in the upward direction.

i.e.  $F = 2.4 \times 10^5 \text{ N}$ , upwards.

Hence, option (3) is the correct answer.

**Q17) Explanation:**

**Hint:**  $F_{\text{bottom}} = mg - F_B$

**Step 1:** Find the volume of the object.

As the object just floats in the first liquid,  $d_o = d_{l_1} = 2.5 \text{ g/ml}$

The volume of the object,  $V_o = \frac{M}{d_1} = \frac{50}{2.5} = 20 \text{ ml}$

**Step 2:** Find the force on the bottom of the container.

The net force on the bottom is given by;

$$F_{\text{net}} = mg - F_B \quad [F_B = \text{Buoyancy force}]$$

$$\Rightarrow F_{\text{net}} = 20 \times (2.5 - 2) \times 10$$

$$\Rightarrow F_{\text{net}} = 100 \text{ g-m/s}^2$$

$$\Rightarrow F_{\text{net}} = 0.1 \text{ kg-m/s}^2 = 0.1 \text{ N}$$

Hence, option (1) is the correct answer.

**Q18) Explanation:**

**Hint:**  $Y = \frac{FL}{\Delta L}$

**Step:** Find the Young modulus of the wire.

The Young modulus of the wire is given by;  $Y = \frac{FL}{\Delta L}$

The elongation in the wire is given as;  $\Delta L = \frac{FL}{AY}$

The elongation in the first wire is given as;

$$\Delta L_1 = \frac{FL}{AY_1} \dots (1)$$

The elongation in the second wire is given as;

$$\Delta L_2 = \frac{FL}{AY_2} \dots (2)$$

The elongation of the complete wire is;  $\Delta L = \Delta L_1 + \Delta L_2$

$$\Rightarrow \frac{F(2L)}{AY} = \frac{FL}{AY_1} + \frac{FL}{AY_2} \Rightarrow \frac{2}{Y} = \frac{1}{Y_1} + \frac{1}{Y_2}$$

Therefore, the Young modulus of the wire is  $\frac{2}{Y} = \frac{1}{Y_1} + \frac{1}{Y_2}$ .

Hence, option (4) is the correct answer.

**Q19) Explanation:**

**Hint:**  $\tau_{avg} = \left| \frac{\Delta L}{\Delta t} \right|$

**Step:** Find the average torque acting on the flywheel.  
The average torque acting on the flywheel is given by;

$$\begin{aligned}\tau_{avg} &= \frac{\Delta L}{\Delta t} \\ \Rightarrow \tau_{avg} &= \frac{0.8 - 3.2}{1.2} = -\frac{2.4}{1.2} \\ \Rightarrow \tau_{avg} &= -2 \text{ N-m} \\ \Rightarrow |\tau_{avg}| &= 2 \text{ N-m}\end{aligned}$$

Hence, option **(1)** is the correct answer.

**Q20) Correct Option Is b**

**Q21) Explanation:**

**Hint:** MOI of a spherical shell,  $I = \frac{2}{3} MR^2$ .

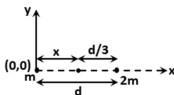
**Step:** Find the relation between the MOI of given objects.

$$\begin{aligned}I_0 &= \frac{2}{3} MR^2 \\ I_S &= \frac{2}{5} MR^2 \\ 5I_S &= 3I_0 \\ \frac{I_S}{3} &= \frac{I_0}{5}\end{aligned}$$

Hence, option (4) is the correct answer.

**Q22) Hint:**  $r_{cm} = \frac{m_1 r_1 + m_2 r_2}{m_1 + m_2}$

Step 1: Draw the diagram.



**Step 2:** Find the position of centre of mass.

$$x_{cm} = \frac{m(0) + 2m(d)}{m + 2m} = \frac{2d}{3}$$

**Step 3:** Find the distance from the heavier mass.

distance from the heavier mass =  $d/3$

**Q23) Explanation:**

**Hint:** The gravitational field inside the sphere varies linearly.

**Step 1:** Find the gravitational field inside the Moon.

The gravitational field inside the Moon is given by;

At  $r \leq R$

$$E = -\frac{GM}{R^3}r$$

$$\Rightarrow E \propto r$$

**Step 2:** Find the gravitational field outside the Moon.

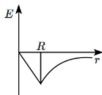
The gravitational field outside the Moon is given by;

At  $r > R$

$$E = -\frac{GM}{r^2}$$

$$\Rightarrow E \propto -1/r^2$$

Therefore, the correct variation of the gravitational field as a function of  $r$  is shown in the figure below;



Hence, option **(2)** is the correct answer.

**Q24) Explanation:**

**Hint:**  $U = Mgh$

**Step:** Find the elastic potential energy stored in the wire.

The potential energy is given by;  $U = Mgh$

$$\Rightarrow U = Mg \frac{l}{2}$$

Hence, option (4) is the correct answer.

**Q25) Explanation:**

**Hint:** Consider earth as a solid sphere.

**Step:** Find the variation of strength of the gravitational field as a function of  $r$ .

The strength of the gravitational field is referred to as intensity and it is given by:

$$E = -\left(\frac{GM}{R^3}\right)r \text{ when } (0 < r \leq R)$$

$$E = -\frac{GM}{r^2} \text{ when } (r \geq R)$$

where,  $M$  is the mass of the earth,  $R$  is the radius of the earth and  $G$  is the gravitational constant.

Therefore, option (1) is correctly representing the variation of a gravitational field as a function of  $r$ .

Hence, option (1) is the correct answer.

**Q26) Explanation:**

**Hint:**  $\eta = \frac{F \times L}{A \times \Delta L}$

**Step:** Find the lateral displacement value of the upper surface w.r.t the lower surface.

The lateral displacement of the upper surface w.r.t the lower surface is given by;

$$\begin{aligned}\Delta L &= \frac{F \times L}{A \times \eta} \\ \Rightarrow x &= \frac{F \times L}{A \times \eta} = \frac{10 \times 10^2 \times 4 \times 10^{-3}}{1 \times 10^{-4} \times 8 \times 10^{11}} \quad [\Delta L = x] \\ \Rightarrow x &= \frac{40}{8} \times 10^{-6} \text{ m} \\ \Rightarrow x &= 5 \times 10^{-6} \text{ m}\end{aligned}$$

Hence, option (2) is the correct answer.

**Q27) Explanation:**

**Hint:**  $t = \sqrt{\frac{2h}{g}}$

**Step:** Find the ratio of time taken by the object to reach the surface of the planet and the earth.

Given:  $g_{\text{earth}} = g$ ,  $g_{\text{planet}} = 3g$

The time taken by the object from a height  $h$  is given by;

$$t = \sqrt{\frac{2h}{g}}$$

The ratio of the time taken by the object to reach the surface of the planet and the earth is given by;

$$\begin{aligned}\frac{t_{\text{planet}}}{t_{\text{earth}}} &= \sqrt{\frac{g_{\text{earth}}}{g_{\text{planet}}}} \\ \Rightarrow \frac{t_{\text{planet}}}{t_{\text{earth}}} &= \sqrt{\frac{g}{3g}} = \frac{1}{\sqrt{3}} \\ \Rightarrow t_{\text{planet}} &= \frac{t_{\text{earth}}}{\sqrt{3}}\end{aligned}$$

Therefore, the time taken by the object to reach the surface of the planet and the earth would be less by a factor of 1.7.

Hence, option (2) is the correct answer.

**Q28) Explanation:**

**Hint:**  $\text{Stress} = \frac{F}{A}$

**Step 1:** Find the stress at end  $A$ .

The tension is maximum at  $A$  since this force is used to carry the whole beam. So, stress at the end  $A$  is maximum.

**Step 2:** Find the stress at end  $B$ .

The tension decreases gradually and becomes zero at  $B$ . So, stress at the end  $B$  is zero.

**Step 3:** Find the stress at end  $C$ .

Stress is directly proportional to the tension of the wire. Therefore, the stress is maximum at  $A$ , intermediate at  $C$  and zero at  $B$ .

Hence, option (2) is the correct answer.

**Q29) Explanation:**

**Hint:**  $g = \frac{GM}{R^2}$

**Step:** Analyse each option one by one.

(1) The value of acceleration due to gravity at a height  $h$  above the ground is given by,

$g_h = g \left[ 1 - \frac{2h}{R_e} \right]$  where  $R_e$  is radius of Earth. Therefore  $g_h$  decreases with an increase in altitude.

(2) The value of acceleration due to gravity at a depth  $d$  below the ground, is given by,

$g_d = g \left[ 1 - \frac{d}{R_e} \right]$  where  $R_e$  is the radius of the earth. So, with an increase in the value of  $d$  the value of  $g_d$  decreases.

(3) The acceleration due to gravity is given by,  $g = \frac{GM}{R^2}$  where  $G$  is the universal gravitational constant,  $M$  is the mass of earth and  $R$  is the radius of the earth. So, the value of  $g$  is dependent on the mass of the earth.

(4) The value of acceleration due to gravity,  $g$  is independent on the mass of the body. Therefore, statement (4) is correct.

Hence, option (4) is the correct answer.

**Q30) Explanation:**

**Hint:** Pressure is defined as force per unit area.

**Step 1:** Find the dimensions of pressure.

Pressure is defined as force per unit area, usually expressed in Pascals (Pa) in the SI system. Its dimensions are:

$$[\text{Pressure}] = \left[ \frac{\text{Force}}{\text{Area}} \right] = \left[ \frac{MLT^{-2}}{L^2} \right] = [ML^{-1}T^{-2}]$$

**Step 2:** Find the dimensions of the elastic modulus.

The elastic modulus generally relates stress (force per unit area) to strain (dimensionless).

$$[\text{Elastic Modulus}] = \left[ \frac{\text{Stress}}{\text{Strain}} \right] = [ML^{-1}T^{-2}]$$

**Step 3:** Find the dimensions of the ratio of pressure to the elastic modulus.

Taking the ratio of pressure to the elastic modulus:

$$\left[ \frac{\text{Pressure}}{\text{Elastic Modulus}} \right] = \left[ \frac{ML^{-1}T^{-2}}{ML^{-1}T^{-2}} \right] = [M^0L^0T^0]$$

Therefore, the ratio of pressure to the elastic modulus is independent of any units and always equals one, making it dimensionless.

Hence, option (4) is the correct answer.

**Q31) Explanation:**

**Hint:** Angular momentum of a planet remains constant.

**Step 1:** Find the kinetic energy of the planet at the perihelion.

The angular momentum is given by;

$$\Rightarrow L = mvr$$

At the perihelion,  $r$  is minimum and  $v$  is maximum. Hence, the kinetic energy of the planet is maximum at the perihelion.

**Step 2:** Compare the time period of the planets which are at different distances from the sun.

As  $T^2 \propto r^3$ , the time period of a planet will be more if it is farther from the sun. Hence, it will take more time to move from the closest position (perihelion) to the farthest position (aphelion).

Hence, option (2) is the correct answer.

**Q32) Explanation:**

**Hint:** Young's modulus depends on the material of the wire.

**Explanation:** Young's modulus is the property of the material, it does not depend up on the dimensions of the material.

Thus, Young's modulus of the wire will remain constant if the length of a wire is made double and the radius is halved of its respective values.

Hence, option (1) is the correct answer.

**Q33) Explanation:**

**Hint:**  $I = mK^2$

**Step:** Find the ratio of its corresponding radii of gyration.

Since no external torque is applied, the angular momentum of the rotating body remains constant i.e.,  $L = I\omega$ .

$$I_1\omega_1 = I_2\omega_2 \dots (1)$$

We can express the moment of inertia in terms of the radius of gyration ( $K$ ) by the expression as;

$$I = mK^2$$

Now we can rewrite the equation (1) in terms of the radius of gyration as;

$$mK_1^2\omega_1 = mK_2^2\omega_2$$

$$\Rightarrow K_1^2\omega_1 = K_2^2\omega_2$$

$$\Rightarrow \frac{K_1^2}{K_2^2} = \frac{\omega_2}{\omega_1}$$

$$\Rightarrow \frac{K_1}{K_2} = \sqrt{\frac{\omega_2}{\omega_1}}$$

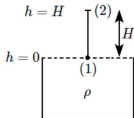
Hence, option (4) is the correct answer.

**Q34) Explanation:**

**Hint:** Gauge Pressure is the amount by which the pressure measured in a fluid exceeds that of the atmosphere.

**Step 1:** Find the water pressure (gauge pressure) at the top of the tank.

Use bernoulli's theorem  $P + \rho gh + \frac{1}{2}\rho v^2 = \text{constant}$



$$P_1 + \rho gh_1 + \frac{1}{2}\rho v_1^2 = P_2 + \rho gh_2 + \frac{1}{2}\rho v_2^2$$

Velocity at point (1) and velocity at point (2) both are zero and height  $h_1 = 0$

$$P = P_{atm} + H\rho g$$

Hence, option (1) is the correct answer.



**Q35) Explanation:**

**Hint:**  $v_E = \sqrt{\frac{2GM}{R}}$

**Step:** Identify the graph between escape velocity vs mass of the earth.

The escape velocity is given by;

$$v_e = \sqrt{\frac{2GM}{R}}$$

$$\Rightarrow v_e \propto \sqrt{M}$$

$$\Rightarrow v_e^2 \propto M$$

The graph is similar to  $y^2 \propto x$ , which represents the parabola as shown in the figure below;



Hence, option **(1)** is the correct answer.

**Q36) Explanation:**

**Hint:**  $Y = \frac{FL}{A\Delta L}$

**Step 1:** Find the ratio for the change in length for steel and copper.

The Young modulus of the material is given by;

$$Y = \frac{FL}{A\Delta L} = \frac{4FL}{\pi D^2 \Delta L}$$

The change in the length of the material is given by;

$$\Delta L = \frac{4FL}{\pi D^2 Y}$$

$$\Rightarrow \frac{\Delta L_S}{\Delta L_C} = \frac{F_S}{F_C} \cdot \frac{L_S}{L_C} \cdot \frac{D_C^2}{D_S^2} \cdot \frac{Y_C}{Y_S}$$

**Step 2:** Find the ratio of increase in their lengths.

$$F_S = (5m + 2m)g = 7mg, F_C = 5mg$$

$$\frac{F_S}{F_C} = \frac{7}{5}, \frac{L_S}{L_C} = q, \frac{D_S}{D_C} = p, \frac{Y_S}{Y_C} = s$$

$$\Rightarrow \frac{\Delta L_S}{\Delta L_C} = \left(\frac{7}{5}\right) (q) \left(\frac{1}{p}\right)^2 \left(\frac{1}{s}\right) = \frac{7q}{5p^2s}$$

Hence, option **(2)** is the correct answer.

**Q37) Explanation:**

**Hint:**  $Y = \frac{\text{Stress}}{\text{Strain}}$

**Step 1:** Find the strain in the given wire.

$$Y = \frac{\text{Stress}}{\text{Strain}}$$

$$2 \times 10^{11} = \frac{8 \times 10^8}{\text{Strain}}$$

$$\Rightarrow \text{Strain} = 4 \times 10^{-3}$$

**Step 2:** Find the maximum elongation in the wire.

$$\text{Strain} = \frac{\Delta L}{L}$$

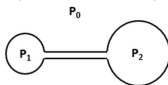
$$\Rightarrow \Delta L = L \times \text{Strain}$$

$$\Rightarrow \Delta L = 4 \times 10^{-3} \text{ m} = 4 \text{ mm}$$

Hence, option **(4)** is the correct answer.

Q38) Hint:  $P \propto \frac{1}{r}$

Step 1: Determine the actual pressure inside two bubbles



$$P_1 - P_0 = \frac{4s}{r_1}$$

$$P_2 - P_0 = \frac{4s}{r_2}$$

$$P_1 > P_2$$

Q39) Explanation:

Hint:  $Y = \frac{F\ell}{A\Delta\ell}$

**Step:** Find the energy density in a wire.

The amount of work done ( $W$ ) in increasing the length of the wire from  $L$  to  $L + l$ , that is from  $l = 0$  to  $l = L$  is;

$$\Rightarrow W = \int_0^L \frac{Y A l}{L} dl = \frac{Y A}{2} \times \frac{l^2}{L}$$

$$\Rightarrow W = \frac{1}{2} \times Y \times \left(\frac{l}{L}\right)^2 \times AL$$

$$\Rightarrow W = \frac{1}{2} \times \text{stress} \times \text{strain} \times \text{volume}$$

The energy density of the wire is  $u = \frac{W}{\text{volume}} = \frac{1}{2} \times \text{stress} \times \text{strain}$

Hence, option (3) is the correct answer.

Q40) Explanation:

**Hint:** The centre-of-mass lies at the midpoint of the chain.

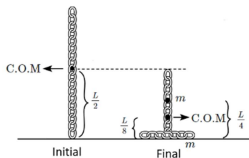
**Step 1:** Find the position of the centre-of-mass.

Initially, the centre-of-mass of the chain is at the centre. When half of the chain has fallen, the centre-of-mass of the hanging part of the chain is at its midpoint, while the centre-of-mass of the portion of the chain lying on the ground is at the bottom. Let the mass of each part be  $m$ . The centre-of-mass of the entire chain is then given by:

$$y_{\text{COM}} = \frac{m\left(\frac{L}{2}\right) + m(0)}{m+m} = \frac{L}{8}$$

When half of the chain has fallen, the centre-of-mass is located at a distance of  $L/8$  from the ground, as shown in the diagram.

The initial and final positions of the centre-of-mass are shown in the figure below:



**Step 2:** Find the shift in the position of the centre-of-mass.

The shift in the position of centre-of-mass is given by;

$$d = \frac{L}{2} - \frac{L}{8}$$

$$\Rightarrow d = \frac{3L}{8}$$

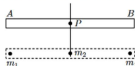
Hence, option (4) is the correct answer.

**Q41) Explanation:**

**Hint:** Use the concept of the moment of inertia of a uniform rod.

**Step 1:** Calculate the moment of inertia of the mass system about  $m_2$ .

Moment of inertia of a rod about point P =  $\frac{mL^2}{12}$ .



Now, we can calculate the moment of inertia of the mass system about  $m_2 = 2(m_1(\frac{L}{4})) = m_1 \frac{L^2}{2}$

According to question

moment of inertia of the mass system about  $m_2$  = Moment of inertia of a rod about point P

$$m_1 \frac{L^2}{2} = \frac{mL^2}{12} \Rightarrow m_1 = \frac{m}{6}$$

**Step 2:** Calculate mass  $m_2$ .

As given in the question,  $2m_1 + m_2 = m, \Rightarrow 2(\frac{m}{6}) + m_2 = m,$

$$m_2 = m - \frac{m}{3} = \frac{2m}{3}$$

**Q42) Explanation:**

**Hint:** The potential energy at a height  $h$  above the earth's surface  $U = \frac{-GMm}{(R+h)}$

**Explanation:** Let's analyze the given statements:

**Assertion (A):** "When a body is raised from the surface of the earth, its potential energy increases."

This is True. The gravitational potential energy of an object increases as it is raised above the surface of the Earth. The potential energy is given by the formula  $U = mgh$ , where  $m$  is the mass of the object,  $g$  is the acceleration due to gravity, and  $h$  is the height above the Earth's surface. As  $h$  increases, the potential energy also increases.

**Reason (R):** "Potential energy of a body on the surface of the earth is zero."

This is False. The potential energy of a body on the surface of the Earth is not necessarily zero. It depends on the reference point chosen. Usually, in problems related to gravitational potential energy, the Earth's surface is taken as the reference point (where potential energy is set to zero), but this is a matter of convention. The absolute potential energy of a body depends on the gravitational field and its distance from the centre of the Earth, not on the reference point chosen.

Therefore, (A) is True and (R) is False.

Hence, option (3) is the correct answer.

**Q43) Explanation:**

**Hint:** The viscous force  $\propto v$ .

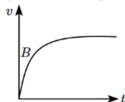
**Step:** Examine the variation of velocity of the spherical ball.

At the moment the ball is released, it starts accelerating under gravity. The velocity increases linearly because the viscous drag is small compared to the gravitational force initially.

As the velocity increases, the viscous drag force (given by Stokes' law,  $F_d = 6\pi\eta rv$ ) also increases. This opposes the motion and reduces the net acceleration.

After some time the viscous drag force and the buoyant force balance the gravitational force, resulting in zero net acceleration. The ball reaches a constant speed called the terminal velocity ( $v_t$ ).

The graph of the velocity of the spherical ball as a function of time is shown in the figure below;



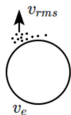
Hence, option **(3)** is the correct answer.

**Q44) Explanation:**

**Hint:** The presence of an atmosphere on any planet implies that  $v_{rms} < v_e$ .

**Explanation:** Analyse the conditions for atmospheric retention. For a planet to retain its atmosphere, the gas molecules must not have enough kinetic energy to escape the gravitational pull. This is mathematically expressed as:

$$v_{rms} < v_e.$$



If  $v_{rms}$  were equal to or greater than  $v_e$ , the gas molecules would possess sufficient speed to escape into space, leading to the loss of the atmosphere.

Therefore, the atmosphere on a planet is only possible if the root mean square speed of gas molecules is less than the escape velocity.

Hence, option **(1)** is the correct answer.

**Q45) Explanation:**

**Hint:**  $\vec{L} = \vec{r} \times \vec{p}$

**Step 1:** Find the linear momentum.

Let  $\vec{r}$  be the position vector of the particle relative to the projection point.

At the highest point of the projectile's motion:

$$\vec{r} = x\hat{i} + y_{\max}\hat{j}$$

where  $x$  is the horizontal displacement, and  $y_{\max}$  is the maximum vertical height.

The velocity of the particle at the highest point is given by;

$$\vec{v} = v_x\hat{i}, \text{ where } v_x = v\cos\theta$$

**Step 2:** Find the angular momentum of the projectile about the point of projection.

$$\vec{L} = \vec{r} \times m\vec{v}$$

$$\Rightarrow \vec{L} = (y_{\max})\hat{j} \times (mv_x)\hat{i}$$

$$\text{where, } y_{\max} = \frac{u^2 \sin^2 \theta}{2g}$$

$$\Rightarrow \vec{L} = -m(v\cos\theta) \left( \frac{u^2 \sin^2 \theta}{2g} \right) \hat{k}$$

$$\text{The magnitude of angular momentum is } L = \frac{mv^2}{2g} \sin^2 \theta \cos \theta = A \sin^2 \theta \cos \theta$$

$$\text{where, } A = \frac{mv^3}{2g}$$

Hence, option **(2)** is the correct answer.

**Q46) In  $\text{NH}_4\text{NO}_3$ , N has two oxidation states i.e. -3 & +5**

*In  $\text{NH}_2\text{OH}$  oxidation of N is -1*

*$\text{NH}_4\text{NO}_3$  exists as  $\text{NH}_4^+$ ,  $\text{NO}_3^-$ ,*

*thus the oxidation state of  $\text{N}^-$  in  $\text{NH}_4^+$  is -3 while in  $\text{NO}_3^-$  is +5.*

*In  $\text{N}_2\text{H}_4$  oxidation of N is -2*

*In  $\text{N}_3\text{H}$  oxidation of N is  $-\frac{1}{3}$ .*

**Q47) Explanation:**

**Hint:** According to 1st law of thermodynamics,  $\Delta U = q + W$

**Explanation:**

Step 1: The given process is isothermal, i.e.,  $\Delta T = 0$

So, the internal energy change is  $\Delta U = C_v \Delta T$ . Thus, the internal energy change will be:

$$\Delta U = 0$$

Step 2: Also, vacuum conditions are given. Hence,  $p_{\text{ext}} = 0$

The work done will be  $(W) = -p_{\text{ext}} \Delta V = 0$

Thus, the heat will be:  $0 = q + 0$ ;  $q = 0$

Therefore, the work done and heat will also be zero. So, **option 3** is the correct answer.

- The gas is expanding **into a vacuum**, meaning there is no external pressure opposing the expansion.
- This is known as **free expansion** (or **Joule expansion**), where the gas expands without doing any work on the surroundings.

**Q48) Explanation:**

Hint: Use the formula that is  $\Delta_r G^\circ = -RT \ln K_p$



If  $N_2O_4$  is 50% dissociated, the mole fraction of both the substance is given by

$$x_{N_2O_4} = \frac{1-0.5}{1+0.5} : x_{NO_2} = \frac{2 \times 0.5}{1+0.5}$$

$$P_{N_2O_4} = \frac{0.5}{1.5} \times 1 \text{ atm}, P_{NO_2} = \frac{1}{1.5} \times 1 \text{ atm}$$

The equilibrium constant  $K_p$  is given by

$$K_p = \frac{(P_{NO_2})^2}{P_{N_2O_4}} = \frac{1.5}{(1.5)^2 (0.5)}$$

$$= 1.33 \text{ atm.}$$

Since

$$\Delta_r G^\circ = -RT \ln K_p, \log 1.33 = 0.1239$$

$$\Delta_r G^\circ = (-8.314 \text{ JK}^{-1} \text{ mol}^{-1}) \times (333K) \times (2.303) \times (0.1239)$$

$$= -789.9 \text{ kJ mol}^{-1}$$

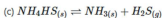
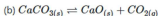
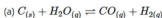
**Q49) Explanation:**

(1) A homogeneous equilibrium is a state where all the reactants and products are in the same phase. The phase can be solid, liquid, or gas.

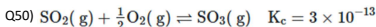
For example,  $2SO_2(g) + O_2(g) \rightleftharpoons 2SO_3(g)$ , all the reactants and products are in gaseous phase.

(2) A heterogeneous equilibrium is one in which species exist in more than one phase. Common examples include reactions involving solids and gases, or solids and liquids.

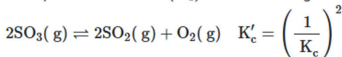
Some examples of heterogeneous equilibrium are:



So, **option 1** is the correct choice.



The equilibrium constant ( $K'_c$ ) for the following reaction



$$K'_c = \left( \frac{1}{3 \times 10^{-13}} \right)^2 = 1.11 \times 10^{25}$$

$$a = 1.11 \text{ and } b = 25$$

$$a + b = 26.11$$

**Q51) Explanation:**

**Hint:** High reduction potential means more likely to gain electrons.

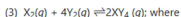
**Explanation:**

According to the electrode potential series, Au ion has a higher tendency to get converted into atomic form. That's why they are known as inert metals.

| S.No. | Electrode    | Reduction Half Reaction         | $E^\circ$ (V) at 298K |
|-------|--------------|---------------------------------|-----------------------|
| 1     | $Au^{3+} Au$ | $Au^{3+} + 3e^- \rightarrow Au$ | +1.50                 |
| 2     | $Fe^{2+} Fe$ | $Fe^{2+} + 2e^- \rightarrow Fe$ | -0.44                 |
| 3     | $Mg^{2+} Mg$ | $Mg^{2+} + 2e^- \rightarrow Mg$ | -2.38                 |
| 4     | $K^+ K$      | $K^+ + 1e^- \rightarrow K$      | -2.95                 |

Thus, the reaction  $Au^{3+} + 3e^- \rightarrow Au_{(s)}$  has the topmost position in EMF series because of maximum SRP.

Hence, **option 3** is the correct answer.

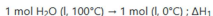
**Q52) Explanation:**

$$\Delta H < 0 \text{ and } \Delta n < 0 \quad [\Delta n = n_p - n_R]$$

$\therefore$  The forward reaction is favored at high pressure and low temperature. (According to Le-Chateliers principle)

**Q53) Explanation:**

STEP 1: The enthalpy changes that take place as follows:



Total enthalpy change will be  $(\Delta H) = \Delta H_1 + \Delta H_2$

STEP 2:

Calculate the value of  $\Delta H_1$  and  $\Delta H_2$  as follows:

$$\Delta H_1 = - (18 \times 4.2 \times 100) \text{ J mol}^{-1} = - 7560 \text{ J mol}^{-1} = - 7.56 \text{ kJ mol}^{-1}$$

$$\Delta H_2 = - 6.00 \text{ kJ mol}^{-1}$$

Therefore,

$$\Delta H = - 7.56 \text{ kJ mol}^{-1} + (-6.00 \text{ kJ mol}^{-1}) = -13.56 \text{ kJ mol}^{-1}$$

There is negligible change in the volume during the change from liquid to solid state. Therefore,  $p\Delta V = \Delta n_g RT = 0$

Consider the following formula,  $\Delta H = \Delta U + \Delta n_g RT$

$$\Delta H = \Delta U = - 13.56 \text{ kJ mol}^{-1}$$

So, **option 2** is the correct answer.

**Q54) Explanation:**

**Hint:**  $K_{sp} = [\text{Au}^{3+}] [\text{Cl}^-]^3$

**Explanation:**

$$\text{Molar solubility} = \frac{1.00 \times 10^{-4} \text{ g AuCl}_3 / 303 \text{ g AuCl}_3 \text{ mol}^{-1}}{1 \text{ L}}$$

$$= 3.3 \times 10^{-7} \text{ M}$$

|               |                                                                  |
|---------------|------------------------------------------------------------------|
| Reaction      | $\text{AuCl}_3 \rightleftharpoons \text{Au}^{3+} + 3\text{Cl}^-$ |
| Initial Conc. | Solid      0 M      0 M                                          |
| Change        | -x      +x      +3x                                              |
| Equilibrium   | Solid      +x      +3x                                           |
| Answer        |                                                                  |

$$K_{sp} = [\text{Au}^{3+}][\text{Cl}^-]^3 = (x)(3x)^3$$

$$\text{Since } x = 3.3 \times 10^{-7}$$

$$K_{sp} = 27 S^4 = 27 \times (3.3 \times 10^{-7})^4 = 3.2 \times 10^{-25}$$

Hence, the **fourth option** is correct.

**Q55) Explanation:**

Hint:  $\text{CH}_3\text{COOH}$  and  $\text{CH}_3\text{COONa}$  is an example of acidic buffer

A buffer is a solution that can resist pH change upon the addition of acidic or basic components. It is able to neutralize small amounts of added acid or base, thus maintaining the pH of the solution relatively stable.

The acidic buffer is formed when weak acid and its salt with a strong base is mixed with each other. For example  $\text{CH}_3\text{COOH}$  and  $\text{CH}_3\text{COONa}$ .

Thus, option 1 is the correct answer.

**Q56) Explanation:**

Step 1: A disproportionation reaction is a redox reaction in which one compound of intermediate oxidation state converts to two compounds, one of higher and one of lower oxidation states.

In a neutral aqueous medium,  $\text{MnO}_4^{2-}$  the following reaction takes place:



Step 2: In this reaction, Manganese disproportionates in +4 and +7 oxidation states.

3 moles of  $\text{MnO}_4^{2-}$  form one mole of  $\text{MnO}_2$  and 2 moles of  $\text{MnO}_4^-$

Hence, 1 mole of  $\text{MnO}_4^{2-}$  forms 1/3 moles of  $\text{MnO}_2$  and 2/3 moles of  $\text{MnO}_4^-$

Thus, **option 1** is the correct answer.



**Q57) Explanation:**

**Hint:** Pressure is an extensive property.

**Explanation:**

Intensive property does not depend on the system size or the amount of material in the system but extensive property depends on the system size or the amount of material in the system.

Pressure is intensive because it is a force per unit area. Force of a gas on the wall of a container is an extensive variable but by dividing it by unit area makes it intensive. This is similar to the specific properties such as specific internal energy,  $U$ , and specific enthalpy,  $H$ , which have units of energy per mass.

The volume is an extensive property as it varies with an object's capacity. The volume of any object is its capacity to keep the matter.

Thus, option 2 is the correct answer.

**Q58) Explanation:**

**Hint:** Metal does not show a negative oxidation state.

Let calculate the oxidation state of the bold atom as follows:



$$\text{Cu} = X, \text{O} = -2$$

$$2X + (-2) = 0$$

$$X = 1$$



$$\text{Cl} = Y, \text{O} = -2$$

$$Y + (-2 \times 3) = -1$$

$$Y = -1 + 6$$

$$Y = 5$$



$$\text{K} = 1$$

$$\text{Cr} = Z, \text{O} = -2$$

$$2 \times 1 + 2Z + (-2 \times 7) = 0$$

$$2Z = 12$$

$$Z = +6$$



$$\text{H} = +1$$

$$\text{Au} = U, \text{Cl} = -1$$

$$1 + U + (-1 \times 4) = 0$$

$$U = +3$$

Hence, option first is the answer.

**Q59) Explanation:**

Hint = The oxidizing substance loses electrons in the reaction, and the reducing substance gains electrons.

**Q60) Explanation:**

HINT: Use general expression of solubility product

**Q61) Hint:**  $KO_2$  is a superoxide.

**Explanation:**

(i) The oxidation number of an atom is defined as the charge that an atom appears to have on forming ionic bonds with other heteroatoms.

The oxidation number of potassium remains the same in all the compounds, i.e., +1. The oxidation numbers of oxygen atoms in  $K_2O$ ,  $K_2O_2$  and  $KO_2$ , are -2, -1, and -1/2 respectively.

(ii)  $K_2O$  is a basic oxide. This pale yellow solid is the simplest oxide of potassium. It is a highly reactive compound that is rarely encountered.

$K_2O_2$  is a peroxide and  $KO_2$  is a superoxide, which forms along with  $K_2O$  when potassium reacts with oxygen.

So, the correct answer is **option 2**.

**Q62) Explanation:**

**Hint:** All the reactants and products must be in the gaseous form.

**Explanation:**

For the reaction,  $2SO_3(g) \rightleftharpoons 2SO_2(g) + O_2(g)$

the equilibrium constant will be written as,

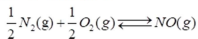
$$K_{eq} = \frac{(P_{SO_2})^2 \cdot P_{O_2}}{(P_{SO_3})^2}$$

All the reactants and products are in the gaseous form.

So, option 4 is the correct answer.

**Q63) Explanation:**

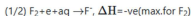
It is a property of equilibrium constant that when an equation is divided by n or multiplied by 1/n,  $K' = K^{1/n}$ . Here, n=2



Therefore,  $K'$  is  $K^{1/2}$

**Q64) Explanation:**

2. Due to smallest halogen, it possesses maximum tendency for accepting electron in aqueous medium.

**Q65) Explanation:**

$$[\text{Salt}] = \frac{n}{v} = \frac{0.0025}{v}$$

$$[\text{Base}] = \frac{n}{v} = \frac{0.15}{v}$$

$$p^{OH} = pK_b + \log \frac{[\text{salt}]}{[\text{Base}]}$$

$$p^{OH} = 4.74 + \log \left[ \frac{0.0025}{0.15} \right]$$

$$= 4.74 - 1.77 = 2.97$$

$$p^H = 14 - p^{OH} = 14 - 2.97 = 11.04$$

**Q66) Hint:  $\Delta G^\circ = -RT \ln K_c$** 

The relation between  $\Delta_r G^\circ$  and  $K_c$  is given below and substitute the value in the given relationship as follows:

$$\begin{aligned} \Delta G^\circ &= -RT \ln K_c \\ &= -8.314 \times 300 \ln 2 \times 10^{13} \end{aligned}$$

**Q67) Explanation:**

HINT: Use general expression for equilibrium constant.

**Q68) Hint:  $\Delta U = q + W$** 

Explanation:

Step 1:

Calculate the value of work done as follows:

$$W = -P \Delta V$$

$$W = -2 \times 10^5 \text{ Nm}^{-2} \times 500 \times 10^{-6} \text{ m}^3$$

$$= -100 \text{ N m}$$

$$W = -100 \text{ J}$$

Step 2:

Calculate the change in internal energy as follows:

$$\Delta U = Q + W$$

$$= 200 - 100$$

$$\Delta U = 100 \text{ J}$$

Q69) Hint:  $\text{KMnO}_4$  changes its colour as the pH of the solution changes

**Because of the colour, the chromate ions  $\text{Cr}^{+3}$  give off at the end of the reaction, potassium dichromate cannot be used as a self-indicator.** The dichromate solutions are usually orange, but they cannot be used as self-indicators.

1. When employed in redox titration, it is reduced to a brown-colored  $\text{Mn}^{2+}$  ion (in acidic solutions) near the endpoint, and the color transformation may be seen fast.
2. A separate indication for permanganate titration is not required because the indicator's function is to identify endpoints.

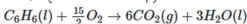
Q70) **Explanation:**

**Hint:** In a combustion reaction, 1 mole of compound reacts with oxygen and carbon dioxide and water is formed.

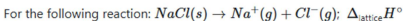
**Explanation:**

(i) The change in enthalpy of a system when one mole of a substance is completely burnt in oxygen or air at a given temperature is known as the enthalpy of combustion.

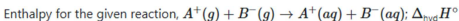
Thus,  $\Delta_c H^\circ$  is the enthalpy of combustion for the reaction:



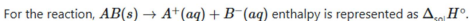
(ii) The lattice formation enthalpy is the enthalpy change when 1 mole of solid crystal is formed from its separated gaseous ions.



(iii) The hydration enthalpy is the enthalpy change when 1 mole of gaseous ions dissolve in sufficient water to give an infinitely dilute solution.



(iv) The enthalpy of solution (or heat of solution) is defined as the change in enthalpy that results when 1 mol of solute (component 1) is dissolved in a solvent (component 2).



Hence, the correct option is first option.

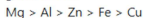
Q71) **Explanation:**

A metal of stronger reducing power displaces another metal of weaker reducing power from its solution of salt.

The order of the increasing reducing power of the given metals is  $\text{Cu} < \text{Fe} < \text{Zn} < \text{Al} < \text{Mg}$ .

Hence, we can say that Mg can displace Al from its salt solution, but Al cannot displace Mg.

Thus, the order in which the given metals displace each other from the solution of their salts is given below:



Q72) **Explanation:**

**Hint:** Equilibrium constant is a measure of the extent or progress to which reaction proceeds.

**Explanation:**

The equilibrium constant of a chemical reaction is the value of its reaction quotient at chemical equilibrium, a state approached by a dynamic chemical system after sufficient time has elapsed at which its composition has no measurable tendency towards further change.

The magnitude of the equilibrium constant  $K$  helps predict the extent to which a reaction proceeds. A large value of  $K_c$  or  $K_p$  shows that forward reaction occurs to a large extent.

Among the given options,  $K = 10^2$  is the highest value. So, this reaction leads the fastest to completion. **Option 4** is the correct answer.

Q73) **Explanation:**

Hint:  $\Delta n_g = (+ve)$

For positive entropy change  $\Delta n_g$  value must be positive.

For first reaction,  $\Delta n_g$  value is negative because aqueous solution is converted into solid form. Solid has less entropy than aqueous solution. Hence, entropy change is negative.

For second reaction,  $\Delta n_g$  value is positive because more gaseous compounds are formed in product side.

For third reaction,  $\Delta n_g$  value is positive because more gaseous compounds are formed in product side as compared to reactant side.

Q74) **Hint:** Cl shows two different oxidation states

The  $\text{CaOCl}_2$  is a bleaching powder. It is written as  $\text{Ca}(\text{OCl})\text{Cl}$ .

The oxidation state of Cl in  $\text{OCl}^-$  is +1 and the Cl directly attached with Ca, oxidation state is -1.

Q75) Explanation:

**Hint:**  $\Delta_r G^\circ = -RT \ln K$

**Explanation:**

STEP 1: We know that,  $\Delta_r G = \Delta_r G^\circ + RT \ln Q$

Also, at equilibrium  $\Delta_r G = 0$  and  $Q = K$

So,  $\Delta_r G^\circ = -RT \ln K$

STEP 2: Given,  $\Delta_r G^\circ = -11.5 \text{ kJ mol}^{-1}$ ;  $T = 300\text{K}$

$-11500 \text{ J mol}^{-1} = -2.303 \times 8.314 \times 300 \times \log K$

or,  $\log K = 2$

$\Rightarrow K = \text{antilog } 2 = 1.0 \times 10^2$

Q76) **Hint:** In  $S_8$ , the oxidation state of sulphur is 0.

**Explanation:**

(i)  $S_8$  contains only one type of atom, which is sulphur. So, as per rule, its oxidation state will be **zero** because the bond contains the same two atoms; hence, there is no electronegativity difference.

(ii) In  $S_2F_2$ , fluorine has a -1 charge. So, the oxidation state of sulphur(x) is  $2x + 2(-1) = 0$  or,  $x = +1$

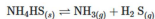
(iii) In  $H_2S$ , hydrogen has a +1 charge. So, the oxidation state of sulphur (x) is  $x + 2(+1) = 0$  or  $x = -2$

Thus, **option 1** is the correct answer.

Q77) Explanation:

**Hint:**  $K_p = P_{NH_3} \times P_{H_2S}$

**Explanation:**



Let initial moles      1                      0                      0

At equilibrium      (1 - x)                      x                      x

Total gaseous moles at equilibrium = x + x = 2x

We known  $K_p = P_{NH_3} \times P_{H_2S}$

but partial pressure (p) = mole fraction  $\times$  total pressure (P)

$$K_p = \left( \frac{x}{2x} \times P \right) \left( \frac{x}{2x} \times P \right) = \left( \frac{P}{2} \right)^2 = \left( \frac{1.12}{2} \right)^2 = 0.3136$$

II<sup>nd</sup> method : Both  $NH_3$  and  $H_2S$  have same number of moles at equilibrium so have same mole fraction and thus equal partial pressures.

ie.,  $P_{NH_3} = P_{H_2S} = \frac{1.12}{2}$

$$K_p = P_{NH_3} \times P_{H_2S} = \frac{1.12}{2} \times \frac{1.12}{2} = 0.3136$$

**Q78) Explanation:**

**Hint:**  $K_p = K_c(RT)^{\Delta n_g}$

**Explanation:**

Step 1: The given reaction is:  $H_2(g) + I_2(g) \rightleftharpoons 2HI(g)$

The relationship between  $K_p$  and  $K_c$  is:  $K_p = K_c(RT)^{\Delta n_g}$

where,  $\Delta n_g$  is the difference between the number of moles of gaseous products and gaseous reactants

Step 2:  $\Delta n_g$  for the given reaction is:  $2 - 2 = 0$

So, the relationship between  $K_p$  and  $K_c$  is:  $K_p = K_c(RT)^0$  or  $K_p = K_c$

For all gaseous reactions, the value of  $K_p$  is not equal to  $K_c$ . It depends on  $\Delta n_g$  of the reaction.

Thus, the assertion is true and the reason is the false statement. **Option 3** is the correct answer.

**Q79) Explanation:**

**Hint:** Enthalpy change,  $\Delta H = \Sigma \text{Bond energies of reactants} - \Sigma \text{Bond energies of products}$

**Explanation:**

(i) The enthalpy change of the reaction is calculated by subtracting the sum of the enthalpies of all the reactants from those of the products. Mathematically,  $\Delta H = \Sigma H_{\text{products}} - \Sigma H_{\text{reactants}}$

This is a correct statement.

(ii) The relationship between enthalpy change and internal energy change is  $\Delta H = \Delta U + P\Delta V$

or  $\Delta H = \Delta U + \Delta nRT$

So, both (2) and (4) are correct.

(iii) In terms of bond energies or enthalpy, the reaction is the difference between the sum of the bond energies of the reactant and product.

$$\Delta H_{\text{(reaction)}}^{\circ} = \Delta H_{\text{(reactant bonds)}}^{\circ} - \Delta H_{\text{(products bonds)}}^{\circ}$$

So, this is not the correct statement. Therefore, **option 3** is the correct choice.

**Q80) Hint:** Hydrogen has a +1 charge.

**Explanation:**

The oxidation number is the hypothetical charge of an atom if all of its bonds to other atoms were fully ionic.

The given molecule is  $\text{CH}_2\text{O}$ . Oxygen has a -2 charge and hydrogen has a +1 charge. Let the oxidation number of a carbon atom be "x." The overall charge on the molecule is 0, as it is a neutral molecule.

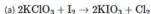
$$x + 2(+1) - 2 = 0 \text{ or } x = 0$$

**Option 3** is the correct answer.

**Q81) Explanation:**

**Step 1:** Oxidation-reduction reactions, commonly known as redox reactions, are reactions that involve the transfer of electrons from one species to another. The species that loses electrons is said to be oxidised, while the species that gains electrons is said to be reduced.

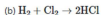
**Step 2:**



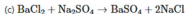
In this reaction, the oxidation states of potassium and oxygen are same on both sides i.e., +1 and -2 respectively. The oxidation state of chlorine changes from +5 (in  $KClO_3$ ) to 0 ( $Cl_2$ ). So, chlorine is getting reduced in the reaction.

Similarly, the oxidation state of iodine changes from 0 ( $I_2$ ) to +5 ( $KIO_3$ ). Hence, iodine is getting oxidised in the reaction. So, it is a redox reaction.

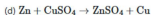
**Step 3:**



Here, hydrogen is getting oxidised from 0 ( $H_2$ ) to +1 ( $HCl$ ) oxidation state and chlorine is getting reduced from 0 ( $Cl_2$ ) to -1 ( $HCl$ ) oxidation state. Thus, it is a redox reaction.



In this case, the oxidation state of all the elements is same on both sides. Thus, it is not a redox reaction.

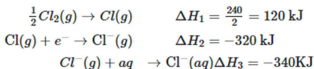


In this reaction, Zn is getting oxidised from 0 to +2 oxidation state and Cu is getting reduced from +2 to 0 oxidation state.

So, the correct answer is **option 3**.

**Q82) Step 1:**

The given values of enthalpies are as follows:



**Step 2:**

Calculate the value of enthalpy for the given reaction is as follows:

$$\begin{aligned} \frac{1}{2}Cl_2(g) + e^- + aq &\rightarrow Cl^-(aq) \\ \Delta H &= \Delta H_1 + \Delta H_2 + \Delta H_3 \\ &= 120 - 320 - 340 \\ &= -540 \text{ kJ} \\ |\Delta H| &= 540 \text{ kJ} \end{aligned}$$

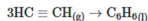
Thus, option first is the correct answer.



Q83) Hint: The relation between  $\Delta G^\circ$  and  $K$  is  $\Delta G^\circ = -RT \ln K$

Step 1:

Calculate the  $\Delta G^\circ$  value for the given reaction



$$\Delta G^\circ = \sum (\Delta G_f^\circ)_p - 3 \sum (\Delta G_f^\circ)_R$$

$$\Delta G^\circ = -1.24 \times 10^5 - (3 \times -2.04 \times 10^5)$$

$$\Delta G^\circ = 4.88 \times 10^5 \text{ J}$$

Step 2:

The relation between  $\Delta G^\circ$  and  $K$  is as follows:

$$\Delta G^\circ = -RT \ln K$$

$$= -RT \ln K = 1 \times (-124 \times 10^5) - (-3 \times 2.04 \times 10^5)$$

$$4.88 \times 10^5 \text{ J} = -R \times T \times \ln K$$

$$\Rightarrow -2.303 \times R \times T \log K = 4.88 \times 10^5$$

$$\Rightarrow \log K = -\frac{4.88 \times 10^5}{2.303 \times R \times T} = -\frac{488000}{5705.848} = 85.52$$

$$= 855 \times 10^{-1}$$

$$\Rightarrow x = 855$$

Q84)

Hint: Isothermal process occur at constant temperature.

1. Adiabatic Process: An adiabatic process is a thermodynamic process which involves the transfer of energy without transfer of heat or mass to the surrounding. In adiabatic process, heat is constant and no transfer of heat takes place.
2. Isolated System: An isolated system is special type of closed system that does not even transfer energy across the boundary.
3. Isothermal Change: The change in which pressure and volume of a gas are changed but the temperature remains constant that change is called isothermal change and the process by which this change occurs is called isothermal process.
4. Path Function: Path functions depend on the path taken to reach one state from another. Heat and work are the example of path function.

Q85) Hint: Species that cannot gain or lose electrons.

Explanation:

**Step 1:** Oxidant: a species that gains electrons, causing another species to be oxidized, and reductant: a species that loses electrons, causing another species to be reduced.

A species can act as both an oxidant and a reductant if it is in an intermediate oxidation state and can either:

- (i) Be oxidized to a higher oxidation state. (ii) Be reduced to a lower oxidation state.

**Step 2:**

$\text{H}_2\text{O}_2$  can act as an oxidizing agent in both acidic and alkaline mediums. It can be oxidized to oxygen  $\text{O}_2$ . Similarly,  $\text{H}_2\text{O}_2$  can act as a reducing agent in both acidic and alkaline mediums, especially when a strong oxidizing agent is present. It can be reduced to water and hydroxide ion.

$\text{SO}_2$  can act as both an oxidizing and reducing agent because sulfur's oxidation state is +4, which can be reduced to +2 or oxidized to +6.

$\text{SO}_3$  can only act as an oxidizing agent because the highest oxidation state of sulfur in  $\text{SO}_3$  is +6.

Nitrous acid ( $\text{HNO}_2$ ) acts as both an oxidant and a reductant. It acts as an oxidizing agent when it changes its oxidation state from +3 to +5. Nitrous acid acts as a reducing agent when it changes its oxidation state from +3 to -3.

Hence, **option 3** is the correct answer.

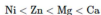
Explanation:

Q86) **Hint:** Metals with more negative  $E^\circ$  values have stronger reducing power.

Explanation:

To determine the reducing power of metals, you would typically arrange them based on their ability to lose electrons (oxidize) and act as reducing agents. Metals with more negative  $E^\circ$  values are stronger reducing agents, while metals with more positive  $E^\circ$  values are weaker reducing agents.

Lower the reduction potential more the tendency to get oxidized and thus more will be reducing power. From the given reduction potential values, the correct order of reducing power is as follows:



Hence, **option 2** is the correct answer.

Explanation:

Q87) (A)  **$\text{CH}_3\text{COONa}$** , sodium acetate, is formed from a strong base (NaOH) and a weak acid ( $\text{CH}_3\text{COOH}$ ). The acetate ion ( $\text{CH}_3\text{COO}^-$ ) is the conjugate base of a weak acid, so it will hydrolyze in water to produce  $\text{OH}^-$  ions, making the solution alkaline. **alkaline, pH > 7.**

(B)  **$\text{NH}_4\text{Cl}$**  Ammonium chloride is formed from a weak base ( $\text{NH}_4\text{OH}$ ) and a strong acid (HCl). The ammonium ion ( $\text{NH}_4^+$ ) is the conjugate acid of a weak base, so it will hydrolyze in water to produce  $\text{H}^+$  ions, making the solution acidic. It is an **acidic salt, pH < 7.**

(C)  **$\text{NaNO}_3$**  sodium nitrate is formed from a strong base (NaOH) and a strong acid ( $\text{HNO}_3$ ). Neither the  $\text{Na}^+$  ion nor the  $\text{NO}_3^-$  ion undergoes significant hydrolysis in water. Thus, it is a **neutral salt, pH = 7.**

(D)  **$\text{CH}_3\text{COONH}_4$**  Dissolution of ammonium ethanoate salt in water results in **pH 7**, but this pH is highly labile. Ammonium acetate does provide buffering around pH 4.75 (the  $\text{pK}_a$  of acetic acid) and around pH 9.25 (the  $\text{pK}_b$  of ammonium).

Hence, **option 3** is the correct answer.

**Hint:**  $\Delta G = \Delta H - T \Delta S$

Q88) For a spontaneous reaction,  $\Delta G$  the value is negative. The  $\Delta G$  value not only depends on  $\Delta H$  value it also depends on the value of  $T$  and  $\Delta S$ . Hence, the correct option is 3, it is difficult to find reaction is spontaneous or non-spontaneous using  $\Delta H$  value.

Q89) Explanation:

**Hint:**  $\sum n_g \neq 0$

Explanation:

**Step 1:** The general relationship between  $\Delta H$  and  $\Delta E$  is as follows:  $\Delta H = \Delta E + \Delta n_g RT$

where,  $\Delta n_g$  is the difference between the number of gaseous products and the number of gaseous reactants

This question aims to find the reaction for which  $\Delta H \neq \Delta E$

**Step 2:** For the given reactions the relationship between  $\Delta H$  and  $\Delta E$  is written as follows:

| S.No. | Reaction                                                                                                           | $\Delta n_g$ | $\Delta H$ and $\Delta E$ relationship |
|-------|--------------------------------------------------------------------------------------------------------------------|--------------|----------------------------------------|
| 1     | $\text{H}_2(\text{g}) + \text{I}_2(\text{g}) \rightarrow 2\text{HI}(\text{g})$                                     | 0            | $\Delta H = \Delta E$                  |
| 2     | $\text{HCl}(\text{aq}) + \text{NaOH}(\text{aq}) \rightarrow \text{NaCl}(\text{aq}) + \text{H}_2\text{O}(\text{l})$ | 0            | $\Delta H = \Delta E$                  |
| 3     | $\text{C}(\text{s}) + \text{O}_2(\text{g}) \rightarrow \text{CO}_2(\text{g})$                                      | 0            | $\Delta H = \Delta E$                  |
| 4     | $\text{N}_2(\text{g}) + 3\text{H}_2(\text{g}) \rightarrow 2\text{NH}_3(\text{g})$                                  | -2           | $\Delta H \neq \Delta E$               |

Hence, **option 4** is the correct answer.

Q90) It states that changes in the temperature, pressure, volume, or concentration of a system will result in predictable and opposing changes in the system in order to achieve a new equilibrium state. From the graph, when hydrogen is added the equilibrium gets disturbed and then a new equilibrium is established. Hence, point C represents re-established equilibrium.





**Mitoprep**

**Sapno Ki Udaan**